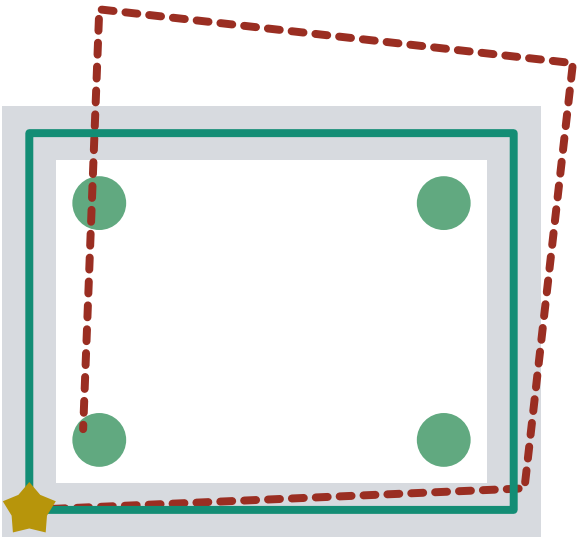
Choosing an IMU: AHRS, GNSS/INS and Tactical GNSS/INS

BIG IDEA

A step-counter alone drifts; “you are here” messages keep you on track.



compass and step-counter only: drifts away

plus “you are here” messages: stays on the block

RTK: give or take 2 cm

THE IDEA

Imagine two friends walking around the block with their eyes closed. The first has a compass and counts steps. The second has the same compass and step-counter but also gets a phone message every second saying “you are about here.” Who gets home closest? With very good messages, the second friend can walk the block and point to the mailbox almost perfectly.

REMEMBER

- ✓ A compass and step-counter alone slowly drift off course.
- ✓ “You are here” messages from GPS stop the drift.
- ✓ Steel, motors and bridges can fool a magnetic compass, so keep it away from them.

TRY THIS

Walk once around a table counting steps, then try it with your eyes closed and a friend calling out “you’re at the corner” at each corner. Which way ends up closer?

The full module is in the Reference Book.